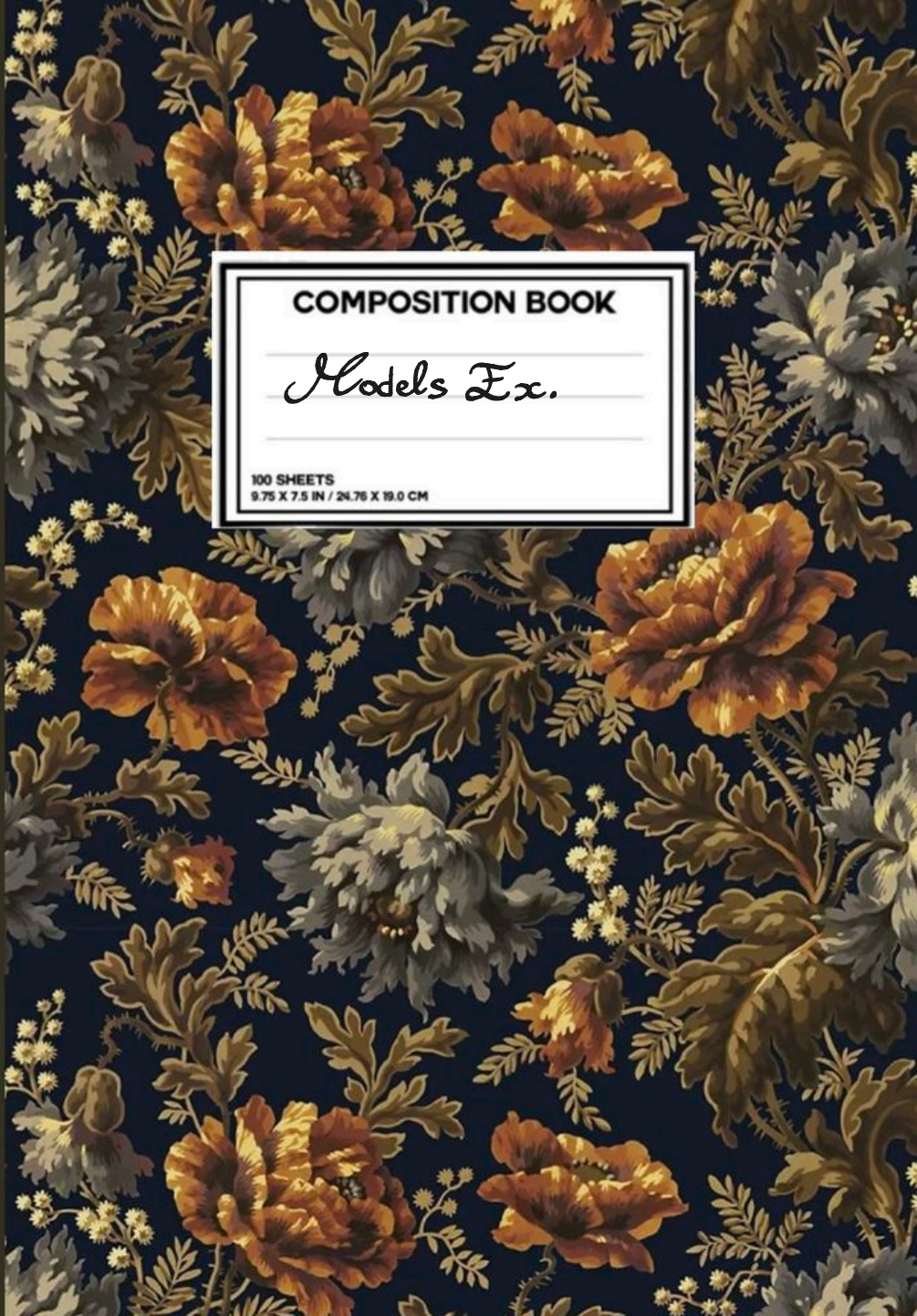

COMPOSITION BOOK
Models Ex.
100 SHEETS
9.75 X 7.5 IN / 24.76 X 19.0 CM

Jan 12. 2017

Ex 1. β-reduce to normal form the term

$$(\lambda xy.x(xy))((\lambda z.zz)y)$$

by means of a call-by-value strategy.

$(\lambda xy.x(xy))((\lambda z.zz)y)$

$(\lambda xw.x(xw))(yy)$

$(\lambda w.(yy)((yy)w)$

Ex 2. Given the abstraction algorithm ($x \notin U, x \in V, W$)

$$\begin{array}{lll} [x] & U\,x & = U \\ [x] & x & = \mathbf{I} \\ [x] & U & = \mathbf{K}\,U \\ [x] & U\,V & = \mathbf{B}\,U\,([x]\,V) \\ [x] & V\,U & = \mathbf{C}\,([x]\,V)\,U \\ [x] & V\,W & = \mathbf{S}\,([x]\,V)([x]\,W) \end{array}$$

where $\mathbf{S}XYZ = XZ(YZ), \mathbf{K}XY = X, \mathbf{I}X = X, \mathbf{B}XYZ = X(YZ), \mathbf{C}XYZ = XZY$, find the term of Combinatory Logic which corresponds to the result of the previous exercise and verify its correctness.

$[w](yy)((yy)w)$

$(yy)w = V$ because $w \in V$

$(yy) = U$ because, $w \notin U$

So we have:

$[w]UV$

and by the rules:

$[w]UV = BU([w]V)$

that is:

$B(yy)([w]((yy)w))$

$yy = U$ because $w \notin U$

Since w is the variable we want deleted, so by the rule:

$[w]Uw = U \rightarrow [w](yy)w = (yy)$

using the result, we have:

$B(yy)(yy) \longrightarrow B(yy)(yy)w \longrightarrow (yy)((yy)w)$

Ex 3. Find a term $\Delta = \lambda x.xUV$ such that

$$\begin{cases} \Delta\Delta &= \mathbf{K} \\ \Delta\mathbf{K} &= \mathbf{S} \end{cases}$$

(hint: you clearly have to find U and V; I suggest to start from the second equation)

The first thing I find is U:

$(\lambda x.xUV)K = S$

$KUV = S$

by the rule $K \equiv \lambda my.m$, so:

$(\lambda my.m)UV = S \rightarrow U = S$

I find V:

$$(\lambda x.xuv)\Delta = K$$

$$\Delta uv = K$$

$$(\lambda x.xuv)uv = K$$

$$uuvv = K$$

replacing with $u = s$

$$ssvv = K$$

by the rule $s = \lambda myz.mz(yz)$, so:

$$(\lambda myz.mz(yz))svv = K$$

$$sv(vv) = K$$

To verify that $sv(vv)$ is equal to K, we apply both to x and y. If they are equal, they will give the same result:

$$Kxy = (\lambda my.m)xy = x$$

so $K = x$

$$sv(vv)xy$$

$$= (\lambda myz.mz(yz))v(vv)xy$$

$$\dots$$

$$= Vx((vv)x)y$$

We have to make sure that is returns x, so V must be:

$$V = \lambda abc.a$$

in fact:

$$(\lambda abc.a)x((vv)x)y \to x$$

so:

$$\Delta = \lambda x.xS(\lambda abc.a)$$

Ex 4. Find two different closed terms with type $(\alpha \to \beta) \to (\beta \to \alpha) \to \beta \to \beta$.

$f : \alpha \to \beta$

$g : \beta \to \alpha$

$y : \beta$

1°

$\lambda fgy.y$ ← WE NEED $\alpha \to \beta$, BUT y IS β

2°

$\lambda fgy.f(gy)$

→ IT'S α BECAUSE g TRANSFORMS INPUT β INTO α $(\beta \to \alpha)$

→ f RECEIVES α AS INPUT, SO THE RESULT IS β $(\alpha \to \beta)$

Ex 5. Assign a simple type to $\lambda xyz.x(xyz)$.

We suppose that:

$y : \alpha$

$z : \beta$

$x : \alpha \to \beta \to \gamma$

so: $xyz = \gamma$, but for the second x, we input γ when it has the value α at the start, so:

$\gamma = \alpha \to x : \alpha \to \beta \to \alpha$

but for the second x, we are passing only one input $x(\overset{1}{xyz})$ instead of two $x\overset{1}{y}\overset{2}{z}$, so it becomes $\beta \to \alpha$:

$$(\alpha \to \beta \to \alpha) \to \alpha \to \beta \to (\beta \to \alpha)$$

June 26 2017

Ex 1. β-reduce to normal form the term

$$\lambda uv.(\lambda z.zz)(\lambda t.tuv).$$

$\lambda uv.(\lambda z.zz)(\lambda t.tuv)$

$\lambda uv.((\lambda t.tuv)(\lambda t.tuv))$

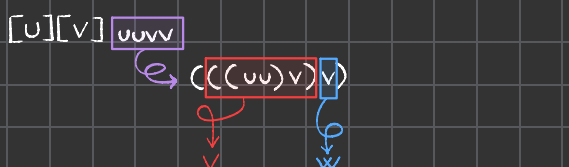
$\lambda uv.((\lambda t.tuv)uv)$

$\lambda uv.uuvv$

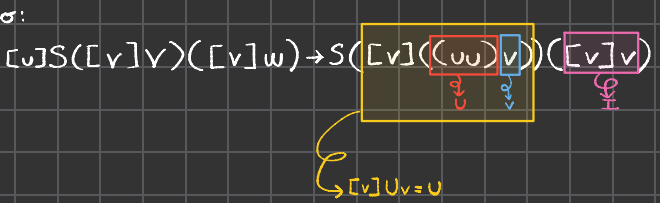
Ex 2. Given the abstraction algorithm ($x \notin U, x \in V, W$)

$$\begin{array}{lll} [x]\ U\ x & = & U \\ [x]\ x & = & \mathbf{I} \\ [x]\ U & = & \mathbf{K}\,U \\ [x]\ U\ V & = & \mathbf{B}\,U\,([x]\,V) \\ [x]\ V\ U & = & \mathbf{C}\,([x]\,V)\,U \\ [x]\ V\ W & = & \mathbf{S}\,([x]\,V)([x]\,W) \end{array}$$

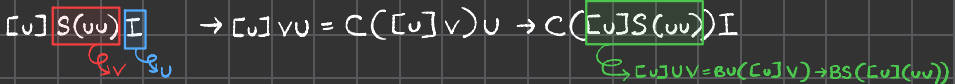
where $\mathbf{S}XYZ = XZ(YZ), \mathbf{K}XY = X, \mathbf{I}X = X, \mathbf{B}XYZ = X(YZ), \mathbf{C}XYZ = XZY$, find the term of Combinatory Logic which corresponds to the result of the previous exercise and verify its correctness.



so:



So we have:



$C(BS([v][uv])I$

$\leadsto vw = S([v]v)([v]w) \to S([v]v)([v]v) \to SII$

$= C(BS(SII))I$

Let's verify that it is correct:

$$C(\underbrace{BS(SII)}_{X})\underbrace{I}_{Y}\underbrace{u}_{Z}v \xrightarrow{Cxyz \to xzy} BS(SII)uIv \xrightarrow{Bxyz \to x(yz)} S(SIIu)Iv \xrightarrow{Sxyz \to xz(yz)}$$

$$\longrightarrow (SIIu)v(\underbrace{Iv}_{Ix=x}) \longrightarrow (SIIu)vv \xrightarrow{Sxyz = xz(yz)} (\underbrace{Iu}_{Ix}(\underbrace{Iu}_{Ix}))vv \longrightarrow uuvv$$

Ex 3. Find a term Δ such that

$$\begin{cases} \Delta(\lambda t.tt) &= y_1 \\ \Delta(\lambda tu.tu) &= y_2 \end{cases}$$

$\Delta = \lambda x.x P_1 P_2 P_3$ | $P_1 = \lambda xy.y$ (a)

$\Delta(\lambda t.tt) = P_1 P_1 P_2 P_3$ | $P_2 = \lambda x.y_1$ (b)

$\Delta(\lambda tu.tu) = P_1 P_2 P_3$ | $P_3 = y_2$ (c)

$\Delta(\lambda t.tt) = P_1 P_1 P_2 P_3 \xrightarrow{(a)} P_2 P_3 \xrightarrow{(b)} y_1$

$\Delta(\lambda tu.tu) = P_1 P_2 P_3 \xrightarrow{(a)} P_3 \xrightarrow{(c)} y_2$

Ex 4. Find two different closed terms with type $(\alpha \to \alpha) \to (\beta \to \alpha) \to \beta \to \alpha$.

$f : (\alpha \to \alpha)$

$g : (\beta \to \alpha)$

$y : \beta$

1°: $\lambda fgy.f(gy)$ 2°: $\lambda fgy.gy$

Ex 5. Assign a simple type to $\lambda xy.xyy$.

$y : \beta$

$x : \alpha$

$xyy : \beta \to \beta \to \alpha$

So:

$(\beta \to \beta \to \alpha) \to \beta \to \alpha$

November 7 2024

Ex 1. β-reduce to normal form the term $(\lambda xyz.xz(yz))(\lambda xyz.x(yz))(\lambda x.x)$.

$$
\begin{aligned}
&(\lambda xyz.xz(yz))(\lambda xyz.x(yz))(\lambda x.x)\\
&=(\lambda yz.(\lambda xyz'.x(yz'))z(yz))(\lambda x.x)\\
&=(\lambda z.(\lambda xyz'.x(yz'))z((\lambda x.x)z))\\
&=(\lambda z.(\lambda yz'.z(yz'))z)\\
&=(\lambda z.(\lambda z'.z(zz')))\\
&=\lambda zz'.z(zz')
\end{aligned}
$$

Ex 2. β-reduce to normal form the term $(\lambda xz.z(xz))((\lambda zy.yz)(\lambda zx.xz)(\lambda u.u)z)$.

$$
\begin{aligned}
&(\lambda xz.z(xz))((\lambda zy.yz)(\lambda zx.xz)(\lambda u.u)z)\\
&=(\lambda z'.z'((\lambda zy.yz)(\lambda zx.xz)(\lambda u.u)zz')\\
&=(\lambda z'.z'((\lambda y.y(\lambda zx.xz))(\lambda u.u)zz')\\
&=(\lambda z'.z'((\lambda u.u)(\lambda zx.xz)zz')\\
&=(\lambda z'.z'((\lambda zx.xz)zz')\\
&=(\lambda z'.z'((\lambda x.xz)z')\\
&=(\lambda z'.z'(z'z))\\
&=\lambda z'.z'(z'z)
\end{aligned}
$$

Ex 3. Find a term X such that $Xxy = Xz(Xy)$

I move X out and replace it with f:

$$
\begin{aligned}
&X=\lambda xy.X_2(Xy)\to(\lambda fxy.fz(fy))X\\
&X=Y(\lambda fxy.fz(fy))
\end{aligned}
$$

Ex 4. Given the abstraction algorithm ($x \notin U, x \in V, W$)

$$
\begin{array}{lll}
[x]\,U\,x = U & [x]\,x = \mathbf{I} & [x]\,U = \mathbf{K}\,U\\
[x]\,U\,V = \mathbf{B}\,U\,([x]\,V) & [x]\,V\,U = \mathbf{C}\,([x]\,V)\,U & [x]\,V\,W = \mathbf{S}\,([x]\,V)([x]\,W)
\end{array}
$$

where $SXYZ = XZ(YZ), KXY = X, IX = X, BXYZ = X(YZ), CXYZ = XZY$, find the term of Combinatory Logic which corresponds to the λ-term $\lambda xy.x(xy)$, and verify its correctness.

$\lambda xy.x(x(xy))$

$=[x][y]x(x(xy))$

$[y]x(x(xy)) \xrightarrow{B\,U([y]V)} Bx([y]x(xy)) \xrightarrow{B\,U([x]V)} Bx(Bx([y](xy)))$, with $Ux = x$

$Bx(Bx\,x)$

$[x]Bx(Bx\,x) \xrightarrow{S([x]V)([x]W)} S([x]Bx)([x]Bx\,x) \xrightarrow{VW = S([x]V)([x]W)} SB(S([x]Bx)([x]x))$, with $Ux = x$

$\to SB(SBI)$

Let's verify that it is correct:

$$SB(SBI)xy \xrightarrow{Sxyz=xz(yz)} Bx((SBI)x)y \xrightarrow{Bxyz=x(yz)} x(SBIxy) \dashrightarrow$$

$$\xrightarrow{Sxyz=xz(yz)} x(Bx(Ix)y) \xrightarrow{Bxyz=x(yz)} x(x(Ixy)) \xrightarrow{Ix=x} x(x(xy))$$

Ex 5. Given $\mathbf{S} \equiv \lambda xyz.xz(yz)$ and $\mathbf{C} \equiv \lambda xyz.xzy$, find a term X such that

$$(\lambda x.xX)\mathbf{C} = \mathbf{S}.$$

$$(\lambda x.xX)C = S$$
$$CX = S$$

Add a and b:

$$Cxab = Sab$$
$$(\lambda xyz.xzy)Xab = Sab$$
$$Xba = Sab$$
$$Xba = (\lambda xyz.xz(yz))ab$$
$$Xba = \lambda z.az(bz)$$
$$X = \lambda baz.az(bz)$$

Ex 6. Given $\mathbf{B} \equiv \lambda xyz.x(yz)$, $\mathbf{K} \equiv \lambda xy.x$ and $\mathbf{C} \equiv \lambda xyz.xzy$, find a term Δ such that $\begin{cases} \Delta\mathbf{B} = y_1 \\ \Delta\mathbf{K} = y_2 \\ \Delta\mathbf{C} = y_3 \end{cases}$

$\Delta = \lambda x.x\ P_1 P_2 P_3 P_4 P_5$

$\Delta B : \lambda xyz.x(yz) = P_1(P_2 P_3) P_4 P_5$

$\Delta K : \lambda xy.x = P_1 P_3 P_4 P_5$

$\Delta C : \lambda xyz.xzy = P_1 P_3 P_2 P_4 P_5$

$P_1 = I$ (a)

$P_3 = \lambda xy.y$ (b)

$P_2 = \lambda xyz.y_1$ (c)

$P_5 = y_2$ (d)

$P_4 = \lambda x.y_3$ (e)

$$\Delta B : P_1(P_2 P_3)P_4 P_5 \xrightarrow{(a)} (P_2 \underset{x}{P_3})\underset{y}{P_4}\underset{z}{P_5} \xrightarrow{(c)} y_1$$

$$\Delta K : P_1 P_3 P_4 P_5 \xrightarrow{(a)} P_3 P_4 P_5 \xrightarrow{(b)} P_5 \xrightarrow{(d)} y_2$$

$$\Delta C : P_1 P_3 P_2 P_4 P_5 \xrightarrow{(a)} P_3 P_2 P_4 P_5 \xrightarrow{(b)} P_4 P_5 \xrightarrow{(e)} y_3$$

2022

Ex 1. β-reduce to normal form the term

$$(\lambda z.z(\lambda y.yx)y)(\lambda xz.(\lambda y.zxy)x).$$

$$(\lambda z.z(\lambda y.yx)y)\overbrace{(\lambda xz.(\lambda y.zxy)x)}^{M}$$
$$= (\lambda z.z(\lambda y.yx)y)(\lambda xz.zxx)$$
$$= M(\lambda y.yx)y$$

$= (\lambda xz.zxx)(\lambda y.yx)y$

$= (\lambda z.z(\lambda y.yx)(\lambda y.yx))y$

$= y(\lambda y.yx)(\lambda y.yx)$

Ex 3. Given the abstraction algorithm ($x \notin U, x \in V, W$)

$$\begin{array}{llll} [x] & U\,x & = & U \\ [x] & x & = & \mathbf{I} \\ [x] & U & = & \mathbf{K}\,U \end{array} \qquad \begin{array}{llll} [x] & U\,V & = & \mathbf{B}\,U\,([x]\,V) \\ [x] & V\,U & = & \mathbf{C}\,([x]\,V)\,U \\ [x] & V\,W & = & \mathbf{S}\,([x]\,V)([x]\,W) \end{array}$$

where $\mathbf{S}XYZ = XZ(YZ), \mathbf{K}XY = X, \mathbf{I}X = X, \mathbf{B}XYZ = X(YZ), \mathbf{C}XYZ = XZY$, produce the term of Combinatory Logic which corresponds to the λ-term

$$(\lambda xy.yx)z(\lambda xyz.xy(zy)),$$

and verify the correctness of the result.

$(\lambda xy.yx)z(\lambda xyz.xy(zy))$

$= (\lambda y.yz)(\lambda xyz.xy(zy))$

$= (\lambda xyz'.xy(z'y))z$

$= (\lambda yz'.zy(z'y))$

$[y][z']\ \underset{U}{zy}\underset{V}{(z'y)} \longrightarrow BU([z']v) = B(zy)([z']z'y) \longrightarrow Bzy(C([z']z')y) \longrightarrow Bzy(CIy)$

$[y]\ \underset{V}{B(zy)}\underset{\omega}{(CIy)} \longrightarrow S([y]B(zy))([y]CIy) \longrightarrow S(BB([y](zy)))(CI) \longrightarrow S(BBz)(CI)$

$S\underset{x}{(BBz)}\underset{Y}{(CI)}\underset{z}{yz'} \longrightarrow (BBzy)((CI)y)z' \longrightarrow B(zy)(CIy)z' \longrightarrow (zy)(CIyz') \longrightarrow (zy)(Iz'y)$

$\longrightarrow zy(z'y)$

Ex 4. Given a term T, find a term X such that $Xyx = Xy(Xx)$

??

$X = \lambda yx.Xy(Xx) \rightarrow (\lambda fyx.fy(fx))$

$X = Y(\lambda fyx.fy(fx))$

Ex 5. Find a term Δ such that $\begin{cases} \Delta(\lambda x.x(\lambda xyz.xz(yz))) = y_1 \\ \Delta(\lambda x.x(\lambda xy.x)) = y_2 \end{cases}$

$\Delta \lambda x.x\,P_1P_2P_3P_4P_5$

$S = (\lambda xyz.xz(yz))$

$K = (\lambda xy.x)$

$\Delta\ \lambda x.x\,SP_1P_2P_3P_4P_5 \rightarrow SP_1P_2P_3P_4P_5 \rightarrow P_2P_4(P_3P_4)P_5$

$\Delta\ \lambda x.x\,KP_1P_2P_3P_4P_5 \rightarrow KP_2P_3P_4P_5 \rightarrow P_2P_4P_5$

$P_2 = \lambda xy.y$ (a)

$P_5 = y_2$ (b)

$P_3 = \lambda xy.y_1$ (c)

$P_1 = P_4 = I$

$\Delta \lambda x.x\,S : P_2P_4(P_3P_4)P_5 \xrightarrow{(a)} (P_3P_4)P_5 \xrightarrow{(c)} y_1$

$\Delta \lambda x.x\,K : P_2P_4P_5 \xrightarrow{(a)} P_5 \xrightarrow{(b)} y_2$